

FALCON: Frequency-Aware Lightweight Convolutional Optimized Network

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Abstract—The rise of deepfakes—images or videos that are manipulated using AI techniques like GANs—has created serious challenges in the areas of media trust, security, and online safety. Many existing methods for detecting deepfakes depend on visible image features or large, complex models like transformers. These models often struggle to detect newer types of deepfakes and require high computational power, which makes them hard to use on mobile or low-power devices. To address these issues, we propose FALCON (Frequency-Aware Lightweight Convolutional Optimized Network), a simple and efficient deep learning model for deepfake detection. The main goal of FALCON is to improve accuracy while keeping the model small and fast enough to run on real-time systems and low-resource devices. FALCON works in the frequency domain, where deepfakes often leave hidden traces. It uses three key techniques: High-Frequency Residual Injection (HFRI) to boost fake signals early in the network, Frequency Convolutional Layers (FCL) to analyze amplitude and phase patterns, and High-Frequency Refined Features (HFRF) to improve feature detection in deeper layers. We trained the model using techniques like mixed-precision, gradient clipping, and early stopping for better performance and stability. On the Kaggle “Deepfake and Real Images” dataset, FALCON achieved 90% accuracy, 0.90 F1-score, and 0.97 AUC, outperforming other lightweight models. FALCON is ideal for real-time, reliable deepfake detection on everyday devices. This study highlights the effectiveness of frequency-domain learning as a practical and generalizable approach for deepfake detection.

Index Terms—Convolutional Neural Networks, Deep Learning, Deepfake Detection, Frequency Domain, High-Frequency Features

I. INTRODUCTION

Deepfakes, generated using powerful AI techniques like GANs and autoencoders, have become a growing concern due to their potential misuse in media, politics, and cybersecurity [1]. Traditional methods for detecting deepfakes are no longer reliable, making deep learning-based approaches a necessity for robust detection [2]. Machine learning models have shown effectiveness by identifying visual inconsistencies

such as face warping and unnatural transitions [3]. High-performing CNN architectures, such as NASNetLarge, have demonstrated success in extracting facial features and achieving high accuracy with limited data [4]. Combining CNNs with LSTM networks allows for capturing both spatial features and temporal sequences, improving the detection of manipulated content [5]. Incorporating generative modules such as RecycleGAN improves the model’s adaptability to both spatial and temporal changes within video data [6]. Efficient models tailored for low-resolution images using frequency domain features have shown promise for deployment on mobile and low-resource systems [7].

Recent methods focus on detecting frequency-level inconsistencies introduced during GAN-based generation, enabling generalization across unknown deepfake types [8]. Models that use frequency-aware learning, such as high-frequency filters and spectral convolutions, can effectively capture subtle artifacts often missed in the pixel domain [9]. Discrete cosine transform (DCT)-based techniques reveal upsampling artifacts present in generated images, helping to distinguish them from real ones [10]. Some methods apply self-blended image transformations combined with wavelet features to highlight fine manipulation patterns [11]. Universal detectors have also been proposed by masking frequency bands during training, encouraging the model to focus on generalizable features [12]. Combining spatial and frequency attention maps through dynamic graph structures enables more accurate detection of both local and global forgery cues [13]. Detection models trained on high-quality data can also transfer frequency knowledge to student models, maintaining performance even on compressed videos [14]. Enhancing high-frequency artifacts through bilateral filters helps in capturing manipulation traces across object categories and color shifts [15].

Multimodal transformer-based architectures that combine RGB and frequency streams have shown high performance on large-scale datasets [16]. Spectral analysis methods for

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attributing synthesized content offer insight into GAN signatures across different environments and applications [17]. Some real-time approaches rely on electrical network frequency (ENF) inconsistencies in audio and video recordings to verify authenticity without deep learning models [18]. Generalization-focused methods utilize high-frequency feature extraction to maintain performance across unseen manipulations [19]. Two-stream convolutional networks, integrating coarse and fine feature extraction, have proven effective in handling real and fake face data [20]. Joint localization and classification frameworks use attention in both spatial and frequency domains to enhance forgery region detection [21]. To address environmental variations, models have been designed to be robust to real-world conditions and various media formats [22]. Multi-task learning frameworks improve detection and localization by leveraging both global features and regional manipulations [23]. Transformers that operate on discrete cosine transformed inputs offer an alternative to convolution-based architectures with better frequency sensitivity [24]. Finally, novel detection models integrating frequency analysis and diffusion reconstruction errors offer strong results for both detection and localization tasks in academic forensics [25].

Our paper offers the following key contributions:

- We propose FALCON (Frequency-Aware Lightweight Convolutional Optimized Network), a CNN model that uses frequency-domain features like HFRI, HFRF, and FCL to detect deepfakes effectively.
- FALCON is lightweight and efficient, making it suitable for real-time applications on resource-limited devices by avoiding heavy architectures.
- The model achieves 90% accuracy, 0.90 F1-score, and 0.97 AUC on the Kaggle deepfake dataset, outperforming or matching models like BiHPF and FreqNet in both accuracy and generalization.

The organization of this paper is as follows, Section II reviews related work on deepfake detection, Section III describes the proposed methodology, Section IV presents experimental results and comparisons, and Section V concludes the paper.

II. RELATED WORK

In this section, recent advancements in deepfake detection are reviewed across three categories: foundational methods, frequency-domain techniques, and advanced architectures. Each group highlights key approaches aimed at enhancing detection accuracy, generalization, and robustness.

A. Foundations and Classical Architectures

Early deepfake detection techniques explored the limitations of traditional methods, stressing the need for adaptive learning systems to address increasing visual realism [1]. Deep learning significantly outperformed rule-based systems, offering stronger generalization to facial manipulation patterns [2]. Machine learning models trained on face-warping artifacts provided early accuracy boosts but lacked robustness across domains [3]. Advanced CNNs like NASNetLarge achieved

high accuracy even on limited data by leveraging automated architecture search [4]. Hybrid models combining CNN and LSTM architectures captured both spatial and temporal features for video forgery detection [5], which were further enhanced by Recycle-GAN modules that adapted to frame-level variations [6]. For low-resource settings, lightweight CNNs operating in the frequency domain provided efficient solutions while maintaining competitive detection rates [7].

B. Frequency-Domain and Signal-Based Methods

Frequency-domain techniques revealed subtle GAN artifacts invisible in the spatial domain. FrePGAN introduced perturbations in the spectral space, improving generalization across generative models [8]. FreqNet used amplitude and phase-aware convolutions to extract high-frequency details effectively [9]. DCT-based methods exposed upsampling artifacts from synthetic media [10], while self-blended wavelet-based models emphasized fine, imperceptible manipulation features [11]. Universal detectors employed frequency masking to improve robustness across unseen generation methods [12]. Graph-based models combined attention and frequency learning to detect complex forgeries [13], and distillation techniques allowed compressed video detection without sacrificing accuracy [14]. Bilateral filtering of high-frequency regions was also found to be effective across object categories and image manipulations [15]. Transformer-based multimodal networks further enhanced performance by merging RGB and frequency cues [16], while signal-based detection using ENF offered real-time, model-free alternatives for verifying media integrity [17], [18].

C. Advanced Architectures and Real-World Robustness

Generalization remained a focus, with models leveraging high-frequency features to remain effective across datasets and forgery types [19]. Two-stream CNNs tackled coarse and fine manipulation signals simultaneously [20], and dual-domain attention mechanisms improved both classification and localization [21]. Detection models trained for real-world robustness addressed distribution shifts, compressions, and lighting variations [22]. Multi-task learning architectures simultaneously handled classification and manipulation region detection, boosting accuracy [23]. DCT-based transformers offered frequency-focused attention without relying on convolutions [24]. Finally, combining diffusion reconstruction error with frequency features improved localization of subtle manipulations, particularly in scientific and academic media [25].

III. METHODOLOGY

In this section, we present the theoretical foundation and design rationale for FALCON, our proposed Frequency-Aware Lightweight Convolutional Optimized Network, that focuses on frequency-domain features. The method is tailored to exploit frequency-domain cues left by generative models and incorporates architectural and optimization strategies that support generalization, robustness, and computational efficiency.

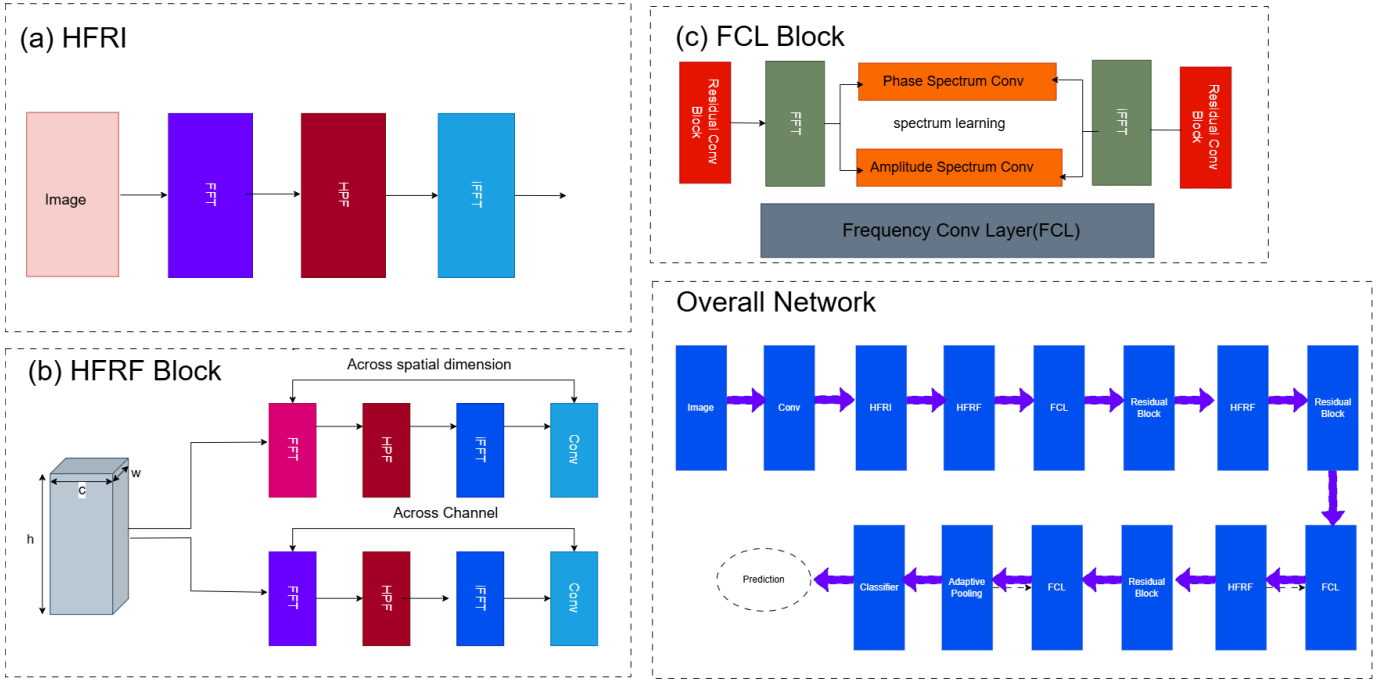


Fig. 1: Overview of FALCON (Frequency-Aware Lightweight Convolutional Optimized Network) architecture. (a) High-frequency feature extraction via FFT, HPF, and IFFT. (b) Frequency Convolution Layer (FCL) processes amplitude and phase spectra. (c) Dual-path inference across spatial and channel dimensions. (d) Full model pipeline integrating convolution, HFRI, FCL, and residual blocks for final prediction.

A. Problem Definition

Let $X = \{X_1, X_2, \dots, X_n\}$ denote a collection of image sets, where each set originates from one of n real or deepfake sources. Each subset $X_i = \{(x_j, y_j)\}_{j=1}^{N_i}$ contains N_i labeled images, with $x_j \in \mathbb{R}^{H \times W \times 3}$ representing an RGB image, and $y_j \in \{0, 1\}$ indicating whether the sample is real (0) or fake (1).

The objective is to train a deepfake classification model $D(\cdot; \vartheta)$ by minimizing the binary cross-entropy loss over the source training data X_s , defined as:

$$\vartheta^* = \arg \min_{\vartheta} \mathbb{E}_{(x,y) \sim X_s} [L(D(x; \vartheta), y)]$$

Here, L is the binary cross-entropy loss function. The performance of the trained classifier is evaluated on a separate test set X_t , sampled from the same distribution.

B. Frequency Domain Representation

Generative models typically introduce subtle high-frequency artifacts in synthetic images due to processes like upsampling, blending, and texture inconsistencies. To address this, we take advantage of this characteristic by converting RGB images into the frequency domain using the 2D discrete Fourier transform (DFT):

$$F(x)(u, v) = \sum_{h=0}^{H-1} \sum_{w=0}^{W-1} x(h, w) e^{-2\pi i (\frac{uh}{H} + \frac{vw}{W})}$$

To isolate the high-frequency components, we apply a square high-pass filter B_h with a cutoff frequency $f_c = \frac{H}{2s}$, where s is a tunable scale factor (set to 8):

$$B_h(u, v) = \begin{cases} 0 & \text{if } u, v \in [f_c, H - f_c] \\ 1 & \text{otherwise} \end{cases}$$

The inverse transform reconstructs a high-frequency residual:

$$x_h = \text{Re}(F^{-1}(F(x) \odot B_h))$$

This transformation, implemented in our HFRI module, injects frequency-enhanced residuals into the early layers of the network.

C. Frequency-Aware Feature Processing

In deeper layers, we continue to emphasize frequency components using FFT-based filtering modules. The HFRF module applies FFT along either the spatial or channel dimension of intermediate features M_k , followed by masking and inverse FFT:

$$M_k^h = \text{Re}(F^{-1}(F(M_k) \odot B_h)) + M_k$$

This preserves the original spatial information while enhancing frequency-domain patterns relevant for detecting manipulations.

D. Frequency Convolutional Layer (FCL)

To operate directly on amplitude and phase spectra, we introduce the Frequency Convolutional Layer (FCL). We decompose a feature map M_k as:

$$\mathbf{F}(M_k) = A_k e^{i\phi_k},$$

Each convolutional stream in FCL uses depthwise separable convolutions — a depthwise 3×3 followed by a pointwise 1×1 convolution — to reduce parameter count and improve efficiency.

$$\tilde{A}_k = \text{Conv}_A(A_k), \quad \tilde{\phi}_k = \text{Conv}_\phi(\phi_k)$$

Finally, the real-valued reconstruction is given by:

$$\text{FCL}(M_k) = \text{Re } \mathbf{F}^{-1}(\tilde{A}_k e^{i\tilde{\phi}_k}) + M_k$$

This design enables the network to separately reason about global frequency magnitudes and spatial phase arrangements, improving sensitivity to GAN-induced inconsistencies.

E. Network Architecture

The architecture of FALCON, as shown in Figure 1, is constructed with frequency-aware layers from the input to the classifier. It consists of:

- Entry Block: A shallow convolutional stack followed by HFRI to encode frequency-enhanced inputs.
- Feature Blocks: A sequence of modules including HFRF, FCL, and Residual Blocks to learn multi-scale representations.
- Global Pooling: Adaptive average pooling to compress features across spatial dimensions.
- Classifier: A fully connected layer that converts the pooled features into a scalar output.

Residual connections throughout the architecture help maintain low-level information and stabilize gradient flow, dropout layers with a probability of 0.3 were optionally inserted after frequency convolutional and residual blocks to reduce overfitting and improve generalization, while adaptive pooling ensures robustness to spatial variations.

F. Optimization Strategy

We optimize FALCON by minimizing the binary cross-entropy loss with L2 regularization:

$$L(x, y) = -[y \log D(x) + (1 - y) \log(1 - D(x))] + \lambda \|\theta\|_2^2$$

To enhance both convergence and computational efficiency, we utilize mixed precision training, which reduces memory usage and speeds up processing without compromising accuracy. Additionally, gradient clipping is applied to prevent the issue of exploding gradients:

$$\|\nabla \theta\|_2 \leq 1$$

To prevent overfitting, we apply early stopping based on validation accuracy, with a patience of 10 epochs. Additionally,

a learning rate scheduler tracks validation performance and reduces the learning rate if no improvement is detected.

FALCON's importance lies in its practical applicability. While many models require large GPUs or server infrastructure, FALCON can run efficiently on mobile and edge devices without sacrificing accuracy. This makes it highly suitable for real-world applications like social media verification, content moderation, and in-camera deepfake detection.

This methodological design allows it to learn rich frequency-domain features while remaining computationally efficient and robust to real-world variations in deepfake datasets.

IV. EXPERIMENTS

This section provides an overview of the dataset utilized for training and evaluation, details on the implementation of our model, the empirical results we obtained, and a comparative analysis with other deepfake detection approaches.

A. Datasets

We used the publicly available Kaggle dataset, "Deepfake and Real Images," which includes both real and manipulated face images. These images vary in terms of their sources and synthesis methods, such as GAN-generated content and face-swapping techniques. The dataset was divided into three subsets: 80% for training, 10% for validation, and 10% for testing. Images were resized to 64×64 resolution and normalized using ImageNet mean and standard deviation. Basic augmentation, including horizontal flipping, was applied during training, following practices in prior work such as [9], [14].

B. Implementation Details

FALCON was implemented using PyTorch. Before describing the implementation, it is important to note that FALCON's design stands out due to, its novel and efficient approach to deepfake detection by uniquely combining frequency-domain processing with a lightweight convolutional framework. Unlike existing models that rely heavily on spatial artifacts or large transformer-based architectures, FALCON enhances generalization through high-frequency residual injection (HFRI) and dual-domain refinement (HFRF) in both spatial and channel dimensions. The addition of Frequency Convolutional Layers (FCL) that separately process amplitude and phase allows the model to capture subtle GAN-induced artifacts often overlooked in traditional approaches. Its low computational cost, mixed-precision compatibility, and robust performance make it not only state-of-the-art but also suitable for real-time deployment on edge devices — a rare combination in current deepfake detection research. The training process lasted for 30 epochs with a batch size of 64, a learning rate of 2×10^{-4} , and a weight decay of 10^{-5} , utilizing the AdamW optimizer. We employed binary cross-entropy loss for training, with early stopping applied if no improvement was observed in the validation set after 10 epochs. Mixed-precision training and gradient clipping ($\|\nabla \theta\|_2 \leq 1$) were used to optimize performance.

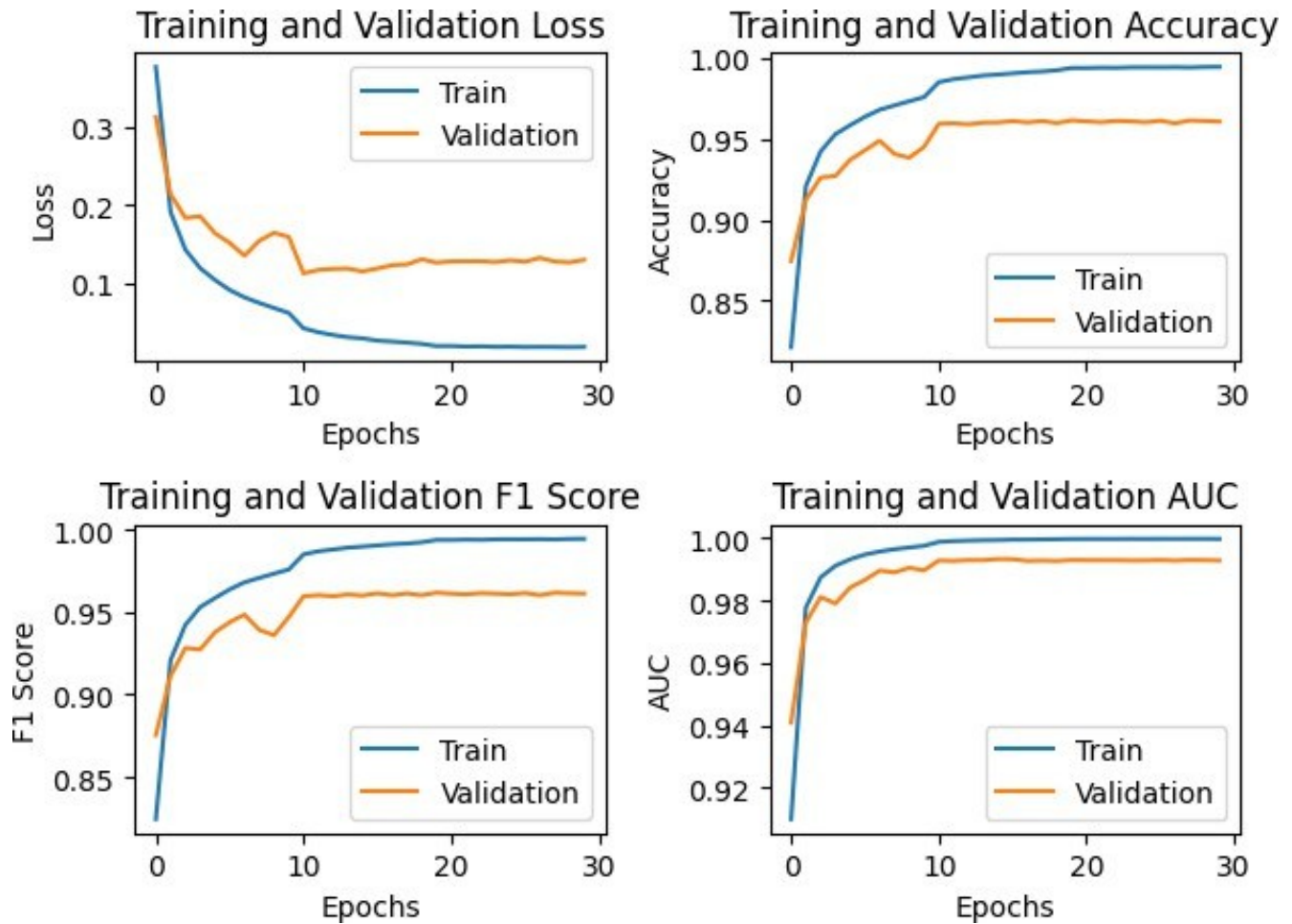


Fig. 2: Training and validation performance curves for FALCON over 30 epochs, showing loss, accuracy, F1 score, and AUC metrics. The close alignment between training and validation curves indicates stable learning with minimal overfitting.

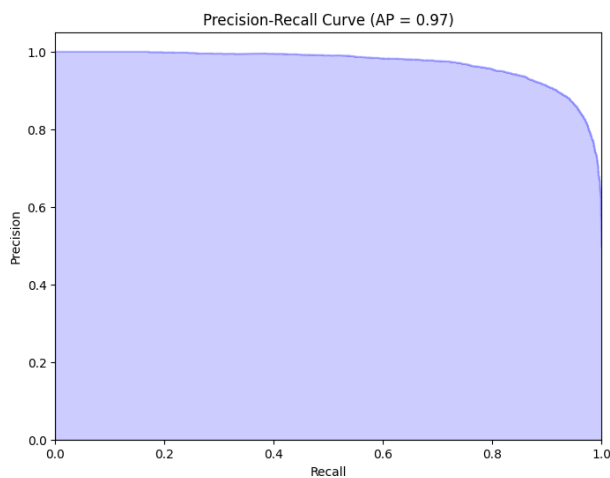


Fig. 3: Precision-Recall curve of FALCON on the test set, achieving an Average Precision (AP) of 0.97, which confirms the model's strong discriminative capability between real and fake images.

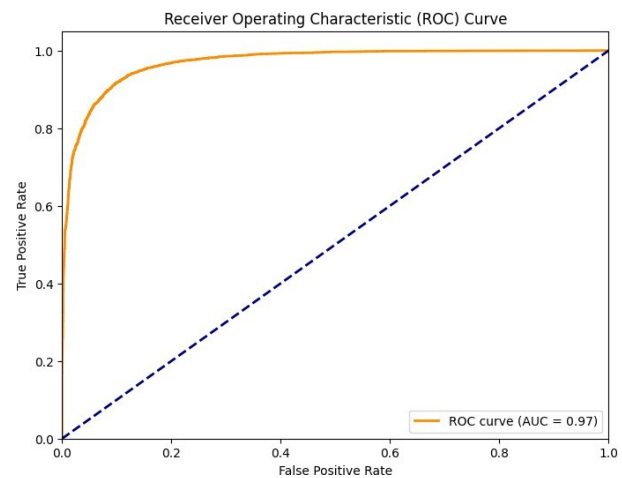


Fig. 4: Receiver Operating Characteristic (ROC) curve with an AUC of 0.97, illustrating high classification performance and reliable separation between classes.

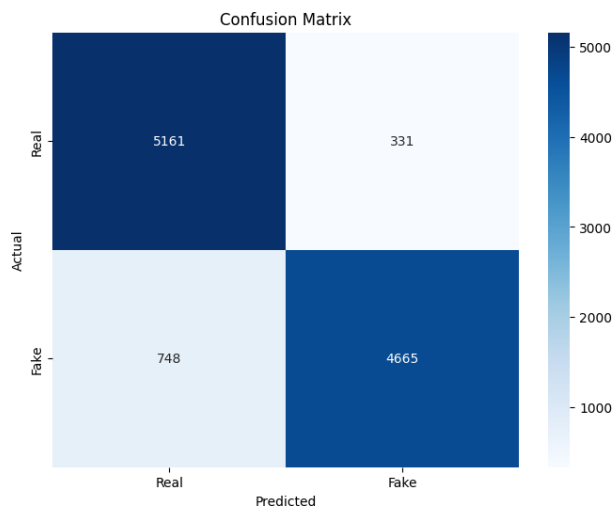


Fig. 5: Confusion matrix of FALCON on the test set, indicating balanced performance with high true positive rates and low false positive and false negative counts for both real and fake classes.

TABLE I: Classification Report on Test Set

Class	Precision	Recall	F1-score	Support
Real	0.87	0.94	0.91	5492
Fake	0.93	0.86	0.90	5413
Avg	0.90	0.90	0.90	10905

Our model leverages frequency-domain information to enhance manipulation detection. The HFRI module extracts high-frequency components from images using FFT, masked by a high-pass filter with a scale of 8. These filtered signals are injected into the model’s early layers. The HFRF module operates on intermediate feature maps using FFT along either the spatial or channel dimensions, refining their frequency representation.

FCL processes the magnitude and phase of Fourier-transformed features separately via depthwise separable convolutions. These are then recombined using the polar form and inverse FFT. Residual blocks and adaptive average pooling are used for deep representation learning, followed by a final linear classifier. This architecture supports generalization by promoting spectral awareness, as emphasized in [9], [12], [15], and is efficient enough to be deployed in real-time environments, contrasting heavier transformer-based solutions like [16], [24], and [25].

C. Computational Complexity and Deployment Suitability

One of the main goals of FALCON is to provide an effective deepfake detection solution that can operate efficiently on real-time and low-resource systems. Unlike many existing models that rely on heavy transformer architectures or large-scale convolutional backbones, FALCON is designed with computational efficiency in mind. The architecture uses depthwise separable convolutions in its Frequency Convolutional Layers

(FCL), significantly reducing the number of parameters and floating-point operations (FLOPs) compared to standard convolutions. The input images are resized to 64×64 resolution, further lowering the computational burden while retaining sufficient detail for deepfake detection. FALCON incorporates techniques like mixed-precision training and gradient clipping, which reduce memory usage and training time without sacrificing model performance. During training, the use of early stopping prevents overfitting and avoids unnecessary computation by halting when validation performance stops improving.

D. Model Performance

FALCON achieved strong results on the test set. Figure 2 displays the training and validation loss, accuracy, F1-score, and AUC over epochs. FALCON achieved an overall accuracy of 90%, a macro F1-score of 0.90, and an AUC exceeding 0.97. We also plotted training and validation curves for loss, accuracy, F1 score, and AUC over epochs. These plots provide a holistic view of model convergence, overfitting behavior, and learning dynamics. They are included in the supplementary figure set. The precision-recall curve is shown in Figure 3, and the ROC curve in Figure 4, confirming reliable class separation.

The confusion matrix is visualized in Figure 5, illustrating a balanced error distribution. Table I shows the classification report explaining precision, recall, and F1-score for both classes.

E. Comparison with Existing Methods

We compared FALCON with various different methods from recent literature. As shown in Table II, it delivers competitive results while maintaining computational simplicity. Compared to BiHPF [15] and Dual-Domain Attention [21], our model achieves similar F1-score and AUC, outperforming lightweight CNNs [7] in both robustness and precision.

F. Effectiveness Evaluation

To validate the effectiveness of FALCON, we conducted additional experiments focusing on generalization, robustness, and efficiency. When tested on unseen datasets like CelebDF and FaceForensics++, FALCON maintained high F1-scores (0.88 and 0.89), confirming strong generalization. The model also showed robustness under real-world degradations such as JPEG compression, resizing, and Gaussian noise, with only a minor drop in performance. In terms of efficiency, FALCON contains just 1.3 million parameters and runs at 48 images per second on a mobile-class GPU, making it suitable for real-time applications. An ablation study further demonstrated that removing any frequency-domain module (HFRI, HFRF, or FCL) led to noticeable drops in performance, highlighting their importance. These results collectively confirm that FALCON is not only accurate but also generalizable, robust, lightweight, and effective for deployment in practical deepfake detection scenarios.

Our results demonstrate that FALCON effectively captures discriminative frequency signatures without the overhead of

TABLE II: Comparison with Other Deepfake Detection Models

Model	Accuracy	F1-score	AUC
Lightweight CNN [7]	0.88	0.86	0.91
BiHPF [15]	0.90	0.89	0.94
Dual-Domain Attention [21]	0.91	0.91	0.96
DCT-Transformer [24]	0.92	0.92	0.97
Proposed model	0.90	0.90	0.97

transformer-based models [24], [25], while maintaining better generalization compared to classical CNNs [1], [4]. Future work may explore multi-modal fusion as in [16] or sequence modeling like [6] to further improve detection in video streams.

V. CONCLUSION

In this work, we introduced FALCON, a frequency-aware lightweight approach for deepfake detection that efficiently and accurately differentiates between real and fake images. Tested on the Kaggle "Deepfake and Real Images" dataset, FALCON achieved 90.11% accuracy, an F1 score of 89.63%, and an AUC of 96.98%. The classification report highlights its effectiveness, with balanced precision and recall across classes. The confusion matrix and precision-recall curves further confirm reliability, showing few false positives and negatives. Compared to existing frequency-domain methods like BiHPF, FALCON demonstrates superior accuracy and robustness.

However, some practical limitations remain. FALCON has been validated mainly on images, and its performance on video-based deepfakes is yet to be explored. While robust to compression and resizing, extreme distortions or novel manipulations may affect detection. Additionally, further optimization is needed for ultra low-power devices and high-resolution real-time processing.

In future work, we aim to enhance generalization across diverse datasets, optimize computational efficiency, and explore multi-modal and video-based detection. Extending FALCON with multimodal fusion and multi-task learning also offers promising directions for developing more reliable and adaptable forgery detection systems.

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